

UNetV4-Akida for Semantic Segmentation on the Cityscapes Dataset

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1 Overview

This report presents a semantic segmentation model based on the UNetV4 architecture with a **MobileNetV4 encoder** and an Akida-compatible lightweight decoder. To balance spatial context and computational constraints, each input image and corresponding mask are **downsampled by a factor of 0.5** prior to training. This effectively doubles the receptive field per pixel, allowing each 384×384 patch to capture a wider scene context while fitting within GPU and neuromorphic deployment constraints.

2 Dataset Preparation

The Cityscapes dataset provides high-resolution urban street scenes with dense pixel annotations for 19 semantic classes. The preprocessing pipeline is adapted to generate downsampled crops that retain large-scale contextual cues.

2.1 Half-Resolution Preprocessing

Each Cityscapes image $I \in \mathbb{R}^{H \times W \times 3}$ and its segmentation label map M are resized by a factor of 0.5 using bilinear interpolation for RGB images and nearest-neighbor interpolation for masks:

$$I' = \text{Resize}(I, 0.5), \quad M' = \text{Resize}(M, 0.5)$$

This preserves scene semantics while enabling larger effective receptive fields within a smaller input tensor.

2.2 Label Mapping

The raw label IDs are remapped to 19 contiguous training IDs via a lookup table (LUT):

$$\text{LUT}[k] = \begin{cases} \text{trainId}(k), & \text{if } k \in \text{valid IDs} \\ 255, & \text{otherwise} \end{cases}$$

where 255 denotes ignored labels in loss computation.

2.3 Crops and Augmentation

Random 384×384 crops are sampled from each downsampled frame. Training augmentations include horizontal flips and color jitter, followed by normalization with ImageNet mean and standard deviation for each random crop. Each training example therefore consists of:

$$(I_c, M_c) \in \mathbb{R}^{384 \times 384 \times 3} \times \mathbb{R}^{384 \times 384}$$

3 Model Architecture

The network follows a UNet-like encoder-decoder design optimized for neuromorphic hardware deployment.

3.1 Encoder

A **MobileNetV4-Small** backbone pre-trained on ImageNet is used as the encoder, extracted in `features_only` mode via the `timm` library. It produces a hierarchy of feature maps:

$$F_i = E(I_c)_i, \quad i \in \{1, 2, 3, 4, 5\}$$

where F_5 corresponds to the deepest bottleneck feature.

3.2 Decoder (Akida-Compatible)

The decoder has been redesigned to comply with Akida’s convolutional hardware constraints. Each stage performs:

- a transposed convolution layer (`ConvTranspose2d`) for $2\times$ upsampling,
- depthwise-separable convolutions (`SepConvBNReLU`) with batch normalization and ReLU,
- skip fusion through concatenation with the corresponding encoder feature map.

This structure preserves UNet-style skip connections while eliminating dense convolutions not supported by Akida neuromorphic accelerators. The final head applies a 1×1 convolution producing logits for 19 classes, followed by nearest-neighbor upsampling to match the crop resolution:

$$\hat{M}_c = \text{Upsample}(\text{Conv}_{1\times 1}(Y_0))$$

4 Training Procedure

4.1 Loss Function with Class Weights

To address the severe class imbalance inherent in Cityscapes, a class-weighted cross-entropy loss is used:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N w_{y_i} \log p_{i,y_i}$$

where w_{y_i} is the normalized weight for class y_i , empirically scaled up to $4\times$ for rare classes such as *motorcycle*, *train*, and *traffic light*. Pixels with label 255 are excluded from the loss.

4.2 Optimization

Training employs the AdamW optimizer with initial learning rate $\eta_{\max} = 5 \times 10^{-4}$ and weight decay 1×10^{-4} . A cosine annealing scheduler decays the learning rate to $\eta_{\min} = 5 \times 10^{-5}$ across 100 epochs, promoting stable convergence. All models are trained with random seed 17 for reproducibility.

4.3 Metrics

Performance is evaluated using mean Intersection-over-Union (mIoU):

$$\text{IoU}_c = \frac{|P_c \cap G_c|}{|P_c \cup G_c|}, \quad \text{mIoU} = \frac{1}{C} \sum_{c=1}^C \text{IoU}_c$$

where P_c and G_c denote predicted and ground-truth pixel sets for class c . Validation predictions are generated via sliding-window inference with and without Hann-window blending.

5 Sliding-Window Inference

Full-resolution validation images are processed using a 384×384 sliding window with 64-pixel overlap. To reduce edge artifacts, Hann blending is applied:

$$W(x, y) = \text{hann}(x) \cdot \text{hann}(y),$$

and accumulated logits are normalized as

$$\hat{Y} = \frac{\sum_{\text{tiles}} W(x, y) \text{logits}(x, y)}{\sum_{\text{tiles}} W(x, y)}.$$

Since the network operates on half-resolution inputs, the resulting prediction tensor $\hat{Y}$ has half the spatial dimensions of the original Cityscapes frame. For quantitative evaluation, these half-resolution logits are **upsampled using bilinear interpolation** to match the full-resolution ground-truth masks prior to computing pixel-wise accuracy and mean Intersection-over-Union (mIoU). This ensures consistency with the official Cityscapes evaluation protocol while preserving hardware feasibility during training and inference.

For faster monitoring, validation is computed on the first five validation images each epoch, while the full validation set is evaluated after training completes.

6 Learning Dynamics

Table 1: Training configuration for the half-resolution UNetV4-Akida model.

Dataset	Cityscapes (19 classes)
Base learning rate	5×10^{-4}
LR scheduler	CosineAnnealing ($5e^{-4} \rightarrow 5e^{-5}$)
Optimizer	AdamW
Weight decay	1×10^{-4}
Loss	Class-weighted cross-entropy (ignore index 255)
Batch size	4
Epochs	100
Tile size	384×384
Overlap between tiles	64 pixels ($\sim 17\%$)
Downsample factor	0.5
Random seed	17
Device	NVIDIA GPU (CUDA)
Backbone	MobileNetV4-Small (ImageNet pretrained)
Decoder	Custom UNetV4 decoder
Frameworks	PyTorch & Albumentations

7 Results

By epoch 100, the Akida-compatible model achieved a validation mean IoU of **61.25%** using plain mosaic inference and **63.08%** with Hann-window blending. The Hann window improved thin-object segmentation consistency and reduced tiling seams, particularly for classes like *bus*, *truck*, and *train*.

Table 2: Per-class Intersection-over-Union (IoU) for the Akida-compatible half-resolution UNetV4 model on the Cityscapes validation set.

Class	IoU (Mosaic)	IoU (Hann)	GT %
road	96.06	96.12	37.65
sidewalk	74.65	74.86	5.41
building	87.64	88.43	21.92
wall	35.04	38.27	0.73
fence	42.48	44.45	0.82
pole	49.88	51.60	1.48
traffic light	47.22	49.51	0.20
traffic sign	59.80	61.35	0.67
vegetation	88.76	89.30	17.32
terrain	48.05	48.31	0.83
sky	91.72	92.15	3.35
person	65.56	65.99	1.30
rider	46.90	46.89	0.22
car	89.94	90.54	6.51
truck	48.67	55.64	0.30
bus	58.89	67.74	0.39
train	34.33	38.42	0.11
motorcycle	32.92	33.65	0.08
bicycle	65.14	65.36	0.71
Mean IoU	61.25	63.08	—

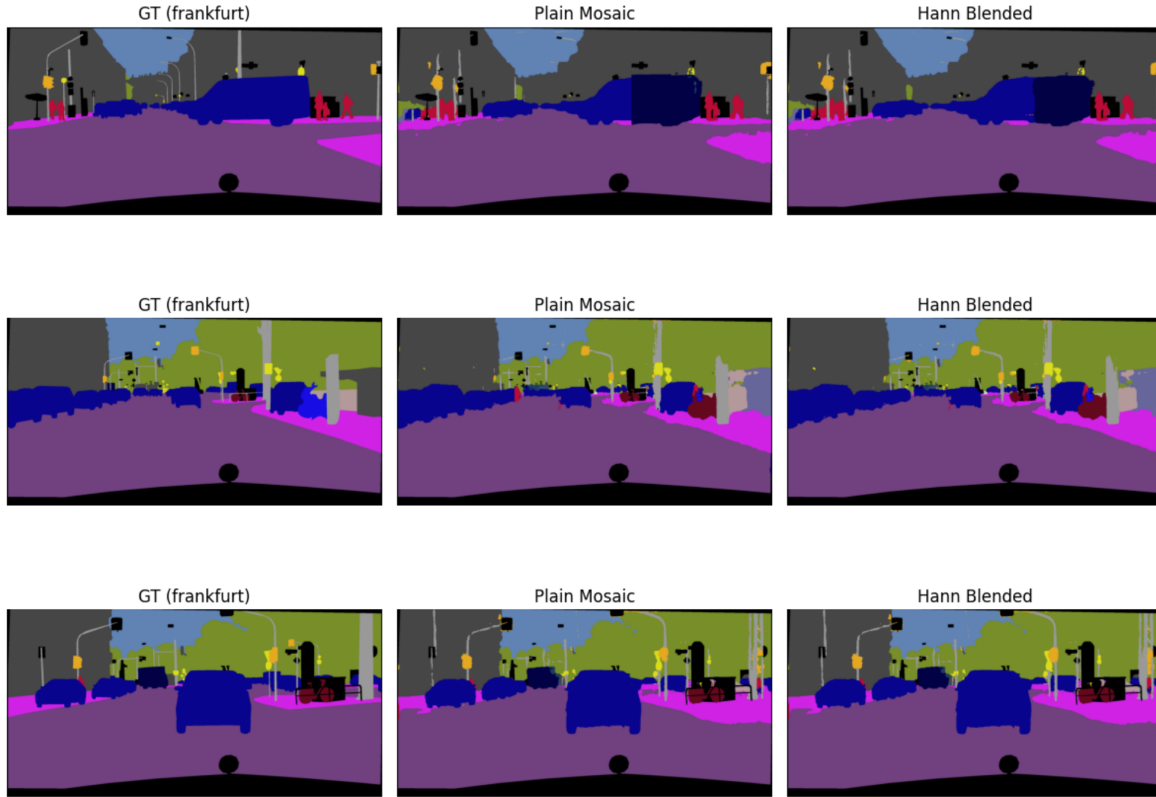


Figure 1: Qualitative comparison on the Cityscapes validation set. Each row shows the ground truth (GT), the model’s prediction using plain non-overlapping tiling (Mosaic), and the same prediction with Hann-window blending. Hann blending reduces visible seams and improves consistency along fine boundaries.

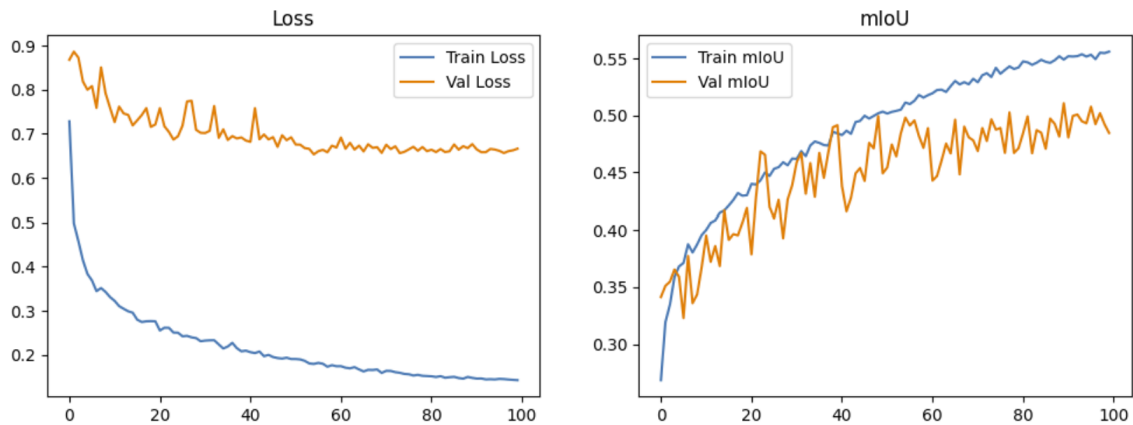


Figure 2: Training and validation performance over 100 epochs. Validation metrics were computed on a subset of five validation images per epoch for faster monitoring, which accounts for oscillations in the validation curves.



Figure 3: Visualization of the tiling layout used during inference. The original image of size 1024×512 is divided into overlapping tiles of 384×384 with 64 pixels overlap along the width, resulting in a 2×3 grid (six total tiles). This layout ensures full spatial coverage while minimizing boundary artifacts.

8 Experiment: Full-Image Downsampling Evaluation

We conducted an additional experiment to test the model’s performance when each Cityscapes validation image was globally downsampled to the same resolution used during training (384×384). The purpose of this experiment was to isolate the effect of resolution loss and evaluate how well the model generalizes when presented with the entire scene at a much lower spatial scale.

8.1 Setup

The model used in this experiment is a **MobileNetV4-UNet decoder** trained for semantic segmentation on Cityscapes. Training was performed for **28 epochs** using **full images uniformly resized to 384×384** . The loss function was cross-entropy with an *ignore index* of 255 to mask out invalid pixels.

During evaluation, each validation image (2048×1024) was resized to 384×384 , processed by the model in a single forward pass, and the resulting logits were **upsampled back to the native resolution** via bilinear interpolation for comparison with the original ground-truth masks. No tiling or overlap blending was applied.

8.2 Results

Table 3 shows the overall mean Intersection-over-Union (mIoU) compared with the half-resolution Hann-blended tiling approach. Despite the $5.3\times$ reduction in input area, the full-image evaluation retained an mIoU of **53.0%**.

The per-class IoU results are detailed in Table 4. Large-scale categories such as *road*, *building*, and *sky* achieved the highest accuracy (above 85%), whereas fine-grained classes like *pole*, *rider*, and *motorcycle* showed reduced scores due to the loss of high-frequency detail during downsampling.

Table 3: Comparison between full-image downsampling and tiling-based evaluation on Cityscapes.

Method	Input Resolution	Evaluation Resolution	mIoU (%)
Tiling (Hann-blended)	1024×512	1024×512	63.1
Full-image downsampling	384×384	384×384	53.0

Table 4: Per-class IoU results for full-image downsampling evaluation.

Class	IoU (%)
road	95.54
sidewalk	70.28
building	84.31
wall	35.90
fence	37.02
pole	32.43
traffic light	29.85
traffic sign	42.48
vegetation	85.33
terrain	51.46
sky	89.69
person	52.25
rider	28.64
car	85.43
truck	35.60
bus	51.50
train	32.51
motorcycle	16.99
bicycle	50.09
Mean IoU	53.02

8.3 Visualization

Qualitative results were generated by comparing the ground-truth masks with predictions produced at both low-resolution and upsampled full-resolution scales. Visual inspection revealed that the model preserved coarse semantic structure across large classes, while small-object boundaries appeared blurred or partially omitted. This indicates that contextual consistency was maintained despite reduced spatial fidelity, demonstrating the model’s robustness to global downsampling.

Comparison of segmentation outputs for a Cityscapes sample. The model prediction at its native training resolution (384×384) and the same prediction upsampled to full resolution are shown alongside the ground truth.

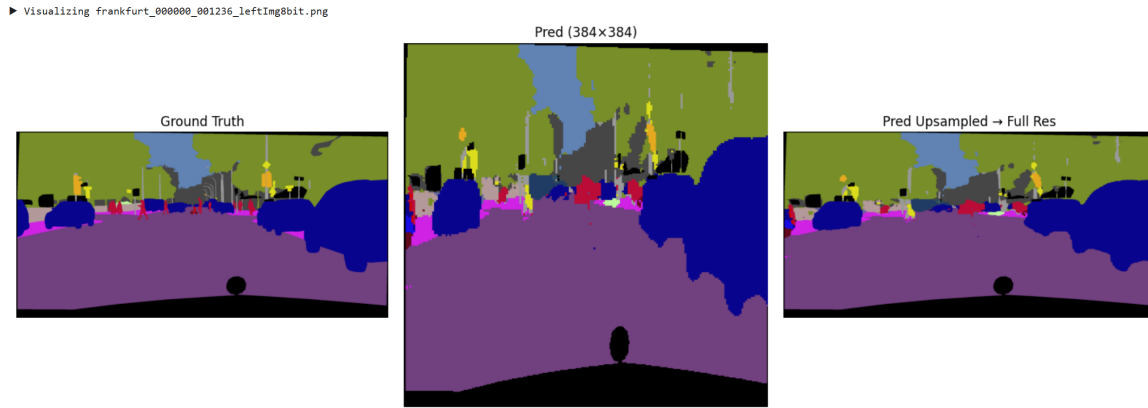


Figure 4: Segmentation comparison for a Cityscapes sample. Left: Ground truth. Center: model prediction at 384×384 resolution. Right: same prediction upscaled to full resolution.

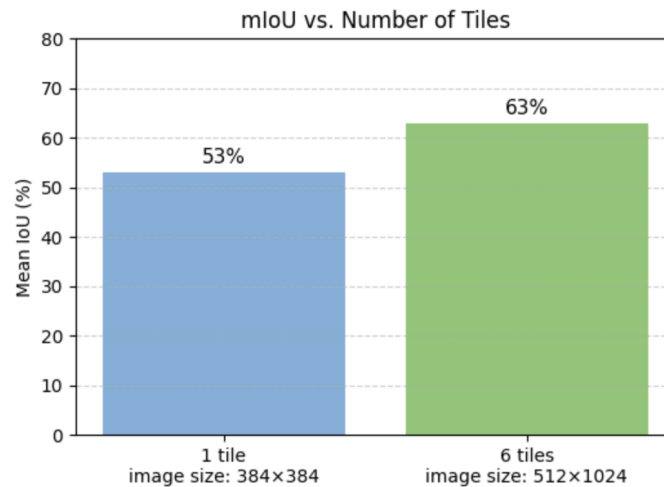


Figure 5: Effect of resolution and tiling on segmentation accuracy. The downsampled full image (384×384 , 1 tile) achieves 53% mIoU, while the half-resolution image (512×1024 , 6 tiles) achieves 63% mIoU.